

Mobile iodides capture for highly photolysis- and reverse-bias-stable perovskite solar cells

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For halide perovskites that are susceptible to photolysis and ion migration, iodide-related defects, such as iodine (I_2) and iodine vacancies, are inevitable. Even a small number of these defects can trigger self-accelerating chemical reactions, posing serious challenges to the durability of perovskite solar cells. Fortunately, before I_2 can damage the perovskites under illumination, they generally diffuse over a long distance. Therefore, detrimental I_2 can be captured by interfacial materials with strong iodide/polyiodide (I_x^-) affinities, such as fullerenes and perfluorodecyl iodide. However, fullerenes in direct contact with perovskites fail to confine I_x^- ions within the perovskite layer but cause detrimental iodine vacancies. Perfluorodecyl iodide, with its directional I_x^- affinity through halogen bonding, can both capture and confine I_x^- . Therefore, inverted perovskite solar cells with over 10 times improved ultraviolet irradiation and thermal-light stabilities (under 85 °C and 1 sun illumination), and 1,000 times improved reverse-bias stability (under ISOS-V ageing tests) have been developed.

Halide perovskites with the formula ABX_3 have created a new era of perovskite materials and devices^{1,2}. The great successes of halide perovskites in optoelectronic devices rely heavily on the use of iodide ions on their X sites, which makes their bandgap narrow enough to absorb (or emit) visible/near-infrared light. Among these, perovskite solar cells (PSCs) are now close to finding practical applications, as witnessed by their very high certified power conversion efficiencies (PCEs)^{3–7}.

However, the instability of PSCs is also closely associated with the presence of iodide ions. This is because, first, the Pb–I bonds in perovskites are prone to break under incident light with $h\nu > 2.5$ eV (that is, photolysis)⁸; and second, I^- ions are mobile in solid-state perovskites even under a mild electrical field of $E \approx 0.2$ V μm^{-1} (refs. 9,10).

In practical use, high-energy photons, local electrical fields and thermal stress dynamically generate iodine vacancy (V_i), iodide interstitial (I_i') and iodine (I_2) defects in perovskite films. V_i are electrically benign but photochemically detrimental because they open up additional decay pathways during long-term operation^{11–14}, such as enhanced superoxide (O_2^-) formation (refs. 13,15). I_2 present in perovskite films can make perovskites decompose more quickly under illumination, mainly due to the formation of highly reactive iodine radicals¹⁶. Iodine radicals trigger the dehydrogenation of organic cations (MA or FA) and generate additional detrimental I_2 molecules, leading to self-accelerating chemical reactions¹⁶. Additionally, the I^- ions migrating out of the perovskite layer can cause irreversible metal corrosion at the electrode

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interface¹⁷, which becomes even more severe under external bias¹⁸. In fact, although the lifetimes of PSCs at maximum power point tracking under 1 sun illumination have been improved to thousands of hours, it has been widely observed that PSCs decay rapidly under a reverse bias of -1.0 to -2.0 V within a few minutes^{19,20}, posing a serious instability issue for these solar devices. In a consensus statement proposed by 59 scientists in 2020, a bias stability (ISOS-V) test was suggested as a necessary protocol¹⁸. Although optimizing the stoichiometric ratio can reduce the initial density of V_i^- or I_i^- (refs. 14, 21), it cannot prevent the dynamic generation of iodide-related defect pairs. Strategies to stabilize these dynamic iodide-related defects and address the relevant device failures are still rare despite their crucial importance.

In this study, a strategy based on iodide/polyiodide capture has been proposed to protect perovskites against dynamically released iodine and irreversible I^- ion loss. We demonstrate the importance and feasibility of maintaining a sustainable halide circulation within the perovskite layer by interfacial iodide/polyiodide capture, which dramatically improves the reverse-bias, ultraviolet and thermal-light stability of the PSCs.

I_2^- and photon-accelerated ageing in perovskites

As a starting point, to gain insights into the detrimental roles of I_2 dissolved in perovskites, extrinsic I_2 molecules were intentionally introduced by spin coating I_2 /isopropyl alcohol (IPA) solution on perovskite surfaces, resulting in I_2 -accelerated perovskite degradation under illumination in a N_2 atmosphere being the dominant failure mechanism (Fig. 1a). As a reference, the bare MAPbI₃ films decompose quickly (see the cross-sectional scanning electron microscopy (SEM) images in Fig. 1b) in the first 10 h of illumination (white light-emitting diode (LED), 100 mW cm⁻², Supplementary Fig. 1), which agrees with previously reported observations¹⁶. Tuning the incident light wavelength, λ (Fig. 1c), reveals that the introduced I_2 molecules are benign when $\lambda > 450$ nm, but become harmful when $\lambda < 450$ nm, which falls into the absorption range of I_3^- (Supplementary Figs. 2–4 and Supplementary Note 1). The much stronger Raman shift peaks at 114 cm⁻¹ (ν_1 mode of I_3^-), 141 cm⁻¹ (ν_3 mode of I_3^-) than that at 169 cm⁻¹ (ν_1 mode of I_2) confirmed that the introduced I_2 mainly bind with I^- and form I_3^- in perovskites^{22,23} (Fig. 1d). This result suggests that the coexistence of photogenerated carriers and I_2 molecules is harmless, but the excitation of I_3^- is detrimental.

The large-size iodine is featured as a polarizable species, which means it is 'soft' according to the hard and soft Lewis acid and base concept²⁴. The electron-accepting I_2 molecule is a kind of soft Lewis acid and can form strong charge-transfer compounds with soft Lewis bases, such as the massive I^- ions in perovskites. I^- can bond with different amounts of iodine to form I_x^- , known as polyiodides and also Lewis bases²⁵. When illuminated by high-energy photons, the excited I_3^- will not necessarily split into I^- and I_2 , but can produce iodine radicals²⁶ ($I^\cdot$ or $I_2^\cdot$, process 1 shown in Fig. 1e):



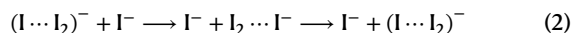
The generated iodine radicals are highly reactive to dehydrogenate organic cations (MA or FA) and trigger the self-accelerating decomposition of MAPbI₃ (Fig. 1b,e).

Suppressed IP-ageing in perovskites with iodides capture

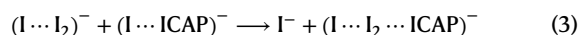
Based on the same IP-ageing test, MAPbI₃ films covered with a few-nanometre-thick coating of Lewis acids, including C₆₀, [6,6]-phenyl-C61-butyric acid methyl ester (PCBM), perfluorodecyl iodide (PFI) and iodopentafluorobenzene (SFIB), or Lewis bases, including trioctylphosphine oxide (TOPO), poly(4-vinylpyridine) (PVP), 3-hexylthiophene (HTP) and 1,8-diiodooctane (DIO), were compared (Supplementary Figs. 5–8). Surprisingly, in contrast to the rapid decomposition of bare MAPbI₃, there were no morphological or optical absorption changes in most of the soft-Lewis-acid-capped MAPbI₃ films after 110 h of

illumination (C₆₀, PCBM, PFI; Fig. 1b and Supplementary Fig. 9). The influence of the Lewis acid capping layer on the penetration of I_2 molecules into MAPbI₃ can be excluded because depositing a C₆₀ layer (2 nm) underneath the MAPbI₃ film can still surprisingly protect it against IP-ageing (Fig. 1b and Supplementary Fig. 10).

The protective effect of the soft Lewis acids over the entire perovskite film indicates that I_2 or iodine radicals may diffuse a surprisingly long distance in perovskite films before they can cause damage. This property enables soft Lewis acids to efficiently scavenge those detrimental I_2 or iodine radicals even deep inside perovskite films (Fig. 1f,g). To prove this, the influence of C₆₀ on the location of I_3^- was studied by Raman shift signal mapping (at 141 cm⁻¹; Fig. 1h and Supplementary Note 2). The samples are MAPbI₃ films with introduced I_2 and a patterned C₆₀ layer. Irrespective of whether the C₆₀ layer was on the top (mapping A) or the bottom of the perovskite layer (mapping B), the I_3^- preferentially accumulated at the regions contacting with the patterned C₆₀. It was also confirmed that the I_3^- ions resulting from the initial I_2 on top of the perovskite films disappeared from the top surface (mapping C) when the C₆₀ layer is located on the bottom side, indicating a robust I_3^- -capture effect of the C₆₀ layer²⁷. The bonding between I_2 and the I^- ions on lattice or grain surface/boundaries may provide the diffusion channels for I_2 (that is, process 3 in Fig. 1f) because the hopping of I_2 between two I^- ions is energetically allowed²⁶:



In the presence of a soft Lewis acid as an iodide/polyiodide capturer (ICAP), the random diffusion of I_2 ends up with their bonding with the ICAP (that is, process 4 in Fig. 1f):



Detailed study confirms that, among several organic electron transport materials, only PCBM and C₆₀ can scavenge I_2 and stabilize the perovskite films efficiently (Fig. 2a), suggesting that the softness of the Lewis acid is more crucial than the lowest unoccupied molecular orbital level of the material (Supplementary Note 3 and Supplementary Figs. 12 and 13). In addition to fullerenes, molecules with fluorocarbon chains and terminated with an iodine atom are also ICAPs. The strong electron-withdrawing properties of fluorocarbon chains make their terminal iodine positively charged and ready to accept the unpaired electron from negatively charged I^- or I_x^- ions. This interaction is also specialized as halogen bonding²⁸. The strength with which the terminal iodine binds with I_x^- is proportional to its positive charge (that is, σ -hole). In our study, PFI and SFIB molecules have been compared in terms of the formation of halogen bonds with iodide compounds (for example, perovskite, formamidinium iodide (FAI), CsI and PbI₂). The chemical shifts of the fluorine atoms on the fluorocarbon chain were characterized by NMR (Fig. 2b,c). After mixing with formamidinium caesium (FACs)-based perovskites, the changes in the chemical shifts of the fluorine atoms nearest to the terminal iodine are significantly larger on PFI (8.1 ppm) than on SFIB (0.4 ppm)^{29,30}. Similarly, PFI forms stronger charge-transfer compounds than SFIB when mixed with FAI, CsI or PbI₂ (Fig. 2c and Supplementary Fig. 14). The shorter separation between electron-withdrawing fluorine and iodine atoms in PFI (I–C–F) compared with in SFIB (I–C–C–F) might be responsible for the higher I^- (or I_x^-) ion affinity of PFI. Consistently, the IP-ageing study shows that the iodide-capture ability of PFI is comparable to that of C₆₀ but much better than that of SFIB (Fig. 2a). According to our density functional theory calculation, the binding energy between the terminal iodine of PFI and the I^- ions on the (001) surface of FA-based perovskite is 0.26 eV with a bond length of 3.38 Å (Supplementary Fig. 15 and Supplementary Note 4).

The interaction between iodide/polyiodide and C₆₀ (and PFI) efficiently suppresses ion migration along the grain surface and boundaries. The measured effective activation energy (E_A) for ion migration (for example, I^-) increases from 0.55 eV to 0.84 eV (or 0.94 eV) after C₆₀ (or

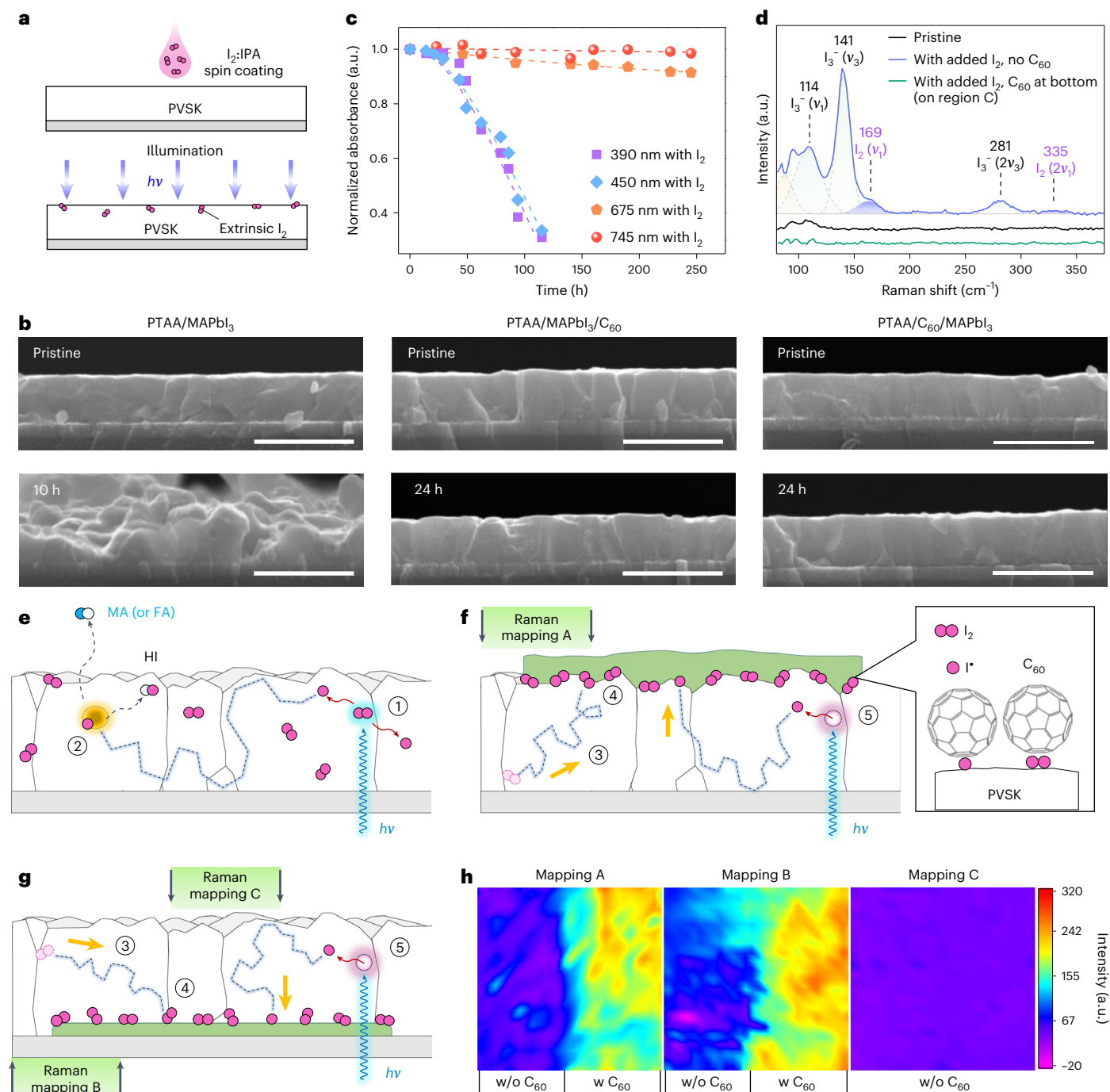


Fig. 1 | Accelerated ageing of perovskites with extrinsic I_2 and photons (IP-ageing) and its suppression. a, The introduction of I_2 in perovskite films for the accelerated ageing test under illumination. **b**, Cross-sectional SEM images of bare PTAA/MAPbI₃, PTAA/MAPbI₃/C₆₀ (10 nm) and PTAA/C₆₀ (2 nm)/MAPbI₃ samples during the IP-ageing process, respectively. Scale bars, 1 μ m. **c**, Influence of the wavelength of incident photons on the decay rate of MAPbI₃ during IP-ageing. **d**, Raman spectra of MAPbI₃ films without (black) or with intentionally introduced I_2 (blue). **e**, Illustration of the IP-ageing of perovskites with introduced I_2 : release of iodine radical from I_2 under illumination (1); iodine-radical-induced

dehydrogenation (2). **f**, Illustration of the long-distance diffusion of I_2 in perovskite films (3), the iodide/polyiodide capture effect (4) of the patterned C₆₀ on the top of perovskite films, and the release of iodine radical from ideal perovskites by photolysis (5). **g**, The iodide/polyiodide capture effect provided by the patterned C₆₀ at the bottom of perovskite film. **h**, Mapping of the Raman shift signal at 141 cm⁻¹ (from I_3^- ions) in region A (marked in **f**, 50 $\times$ 50 μ m²), region B and region C (marked in **g**, 50 $\times$ 50 μ m²), respectively. The Raman spectrum taken at region C is also shown in **d** (green).

PFI/C₆₀) coverage (Fig. 2d–f). Moreover, while there was no absorption change in the FACs-based perovskite sample after a primary ageing of 48 h with I_2 and illumination (Supplementary Fig. 16), the concentration of mobile ions in the control sample increased by ~30 times, and the E_A decreased from 0.55 eV to 0.41 eV, which implies that new defects were released and faster ion-migrating channels were formed at this initial

stage of IP-ageing. By contrast, both C₆₀ and PFI/C₆₀ passivated samples show overlapped Arrhenius plots after the same ageing process, indicating that the IP-ageing has been suppressed by C₆₀ and PFI/C₆₀.

In perovskite films without extrinsic I_2 , the detrimental I_2 can also be generated by other excitations, such as the breaking of Pb–I bonds with high-energy photon irradiation (that is, photolysis at

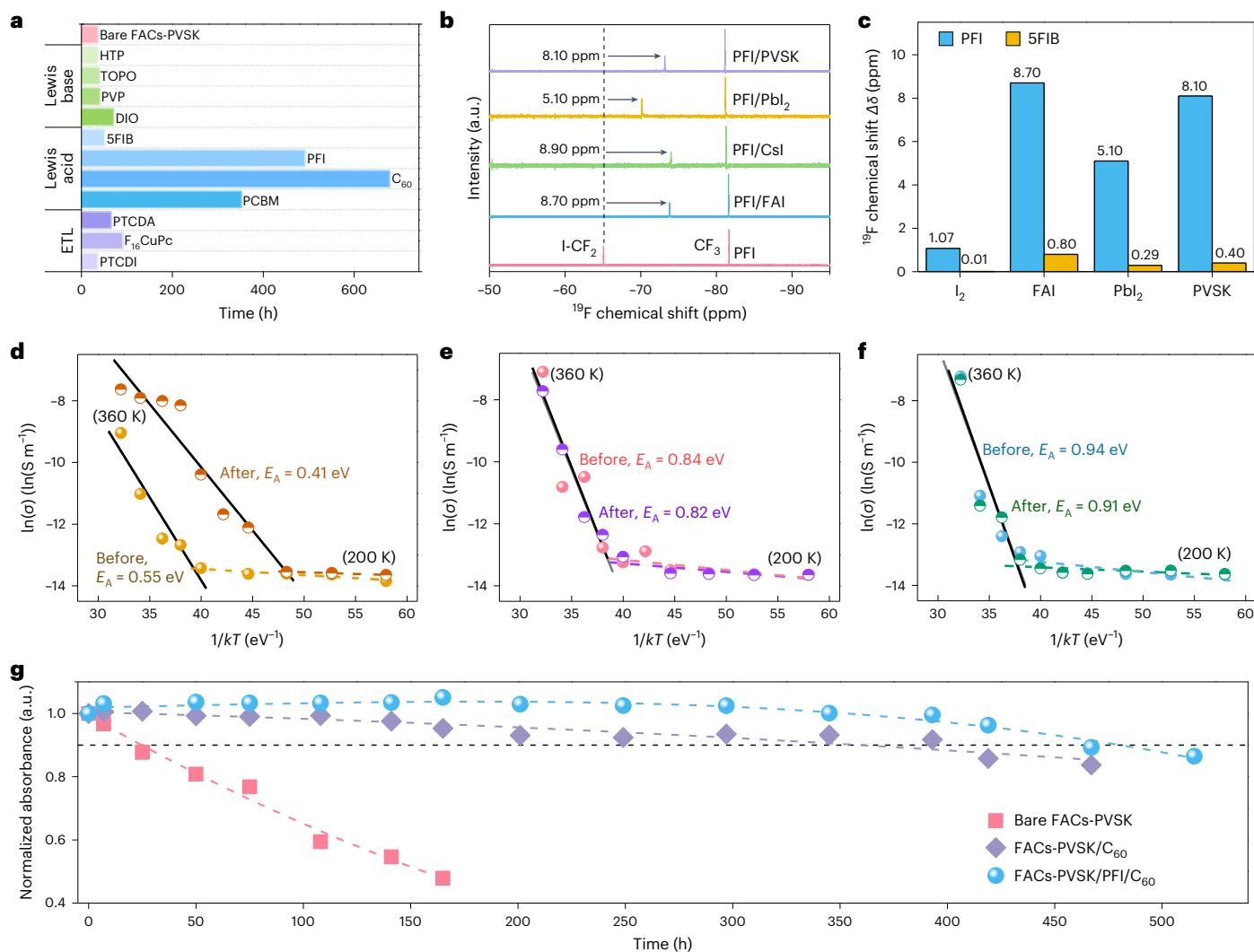


Fig. 2 | Molecules with strong iodide/polyiodide affinity to prevent perovskites from IP-ageing. **a**, Influence of the capping materials on the ageing time needed for FACs-based perovskites to lose 10% of their absorption at a wavelength of 600 nm. **b**, Chemical shifts (δ) of the fluorine atoms on PFI before and after binding with different iodide compounds, respectively. **c**, The changes in chemical shifts ($\Delta\delta$) of the fluorine atoms nearest to the terminal iodine on PFI and 5FIB after binding with iodine or iodide compounds, respectively.

d–f, Arrhenius plots of the conductivity of the bare FACs-PVSK (composition, FA_{0.90}CS_{0.10}PbI_{2.83}Br_{0.17}) (**d**), FACs-PVSK/C₆₀ (**e**) and FACs-PVSK/PFI/C₆₀ (**f**) films before and after 48 h of IP-ageing, respectively, where the E_A are calculated from the slope of the curves. **g**, Absorption reduction (at 600 nm) of FACs-PVSK, FACs-PVSK/C₆₀ and FACs-PVSK/PFI/C₆₀ films (without extrinsic I₂) under ultraviolet light exposure (395 nm, 60 mW cm⁻²) at 85 °C.

$h\nu > 2.5$ eV) or heating³¹. Encouragingly, covering perovskite films with a PFI/C₆₀ (10 nm) bilayer slows down the ultraviolet-induced decomposition process by ~20 times (Fig. 2g and Supplementary Fig. 17). The PFI/C₆₀-covered FACs perovskites do not show any absorption changes after a harsh ultraviolet light aging of 393 h under 60 mW cm⁻² at 85 °C (that is, 235 kWh m⁻²).

ICAPs for PSCs with high reverse-bias stability

The results above consistently proved the notable binding of Γ^-/I_x^- ions with C₆₀ and PFI. Systematic studies proved that PFI can protect different perovskites with varied Br⁻ ion substitution, for example, FA_{0.9}CS_{0.1}Pb(I_{1-x}Br_x)₃ ($x = 0\%$, 5.7%, 10%), against detrimental I₂, indicating the universality of using ICAPs (Supplementary Figs. 18–20). However, it must be emphasized that an ICAP with good iodide affinity can also be a double-edged sword if it promotes the escape of Γ^-/I_x^- ions. For example, a previous study proposed and proved that metals (for example, silver) outside the perovskite layer can scavenge mobile Γ^- ions by forming an irreversible chemical reaction, which gives a reduced current hysteresis but may not be helpful for device stability³².

Unfortunately, C₆₀ with spherical symmetry binds with Γ^-/I_x^- ions non-directionally (Fig. 3a) and hence forms continuous channels for the iodide to transport toward and react with metal cathode. Owing to this feature, Γ^- ions or I₂ molecules have been frequently found in C₆₀ or PCBM layers^{33,34}. The non-directional binding between C₆₀ and Γ^-/I_x^- ions becomes an even more severe problem in reverse-biased PSCs because the applied electrical field aggravates the Γ^-/I_x^- ion migration. In our study, for FACs-based control PSCs with C₆₀ as the electron transport layer (ETL) directly contacting with perovskite, the PCE dropped dramatically from 18.5% to 7.1% (Fig. 3b) after a short reverse-bias treatment (−1.0 V, 200 s); this rate of PCE decline is in agreement with other reported results^{19,20}.

As a major difference, the interaction between PFI and Γ^-/I_x^- ions aided by halogen bonding is directional. The σ -hole located at the end of the terminal iodine only allows Γ^- ions to attach in the opposite direction of the I–C bonds with an I–I–C binding angle of nearly 180°^{28,35} (Fig. 3a). Here, we use this property of PFI to lock the Γ^-/I_x^- ions at the PVSK/PFI interface and meanwhile scavenge harmful I₂ (and iodine radicals) from the perovskite layer. Encouragingly, PSCs with PFI/PCBM/C₆₀ as the ETL become highly stable under reverse-bias stress (Fig. 3b, Supplementary

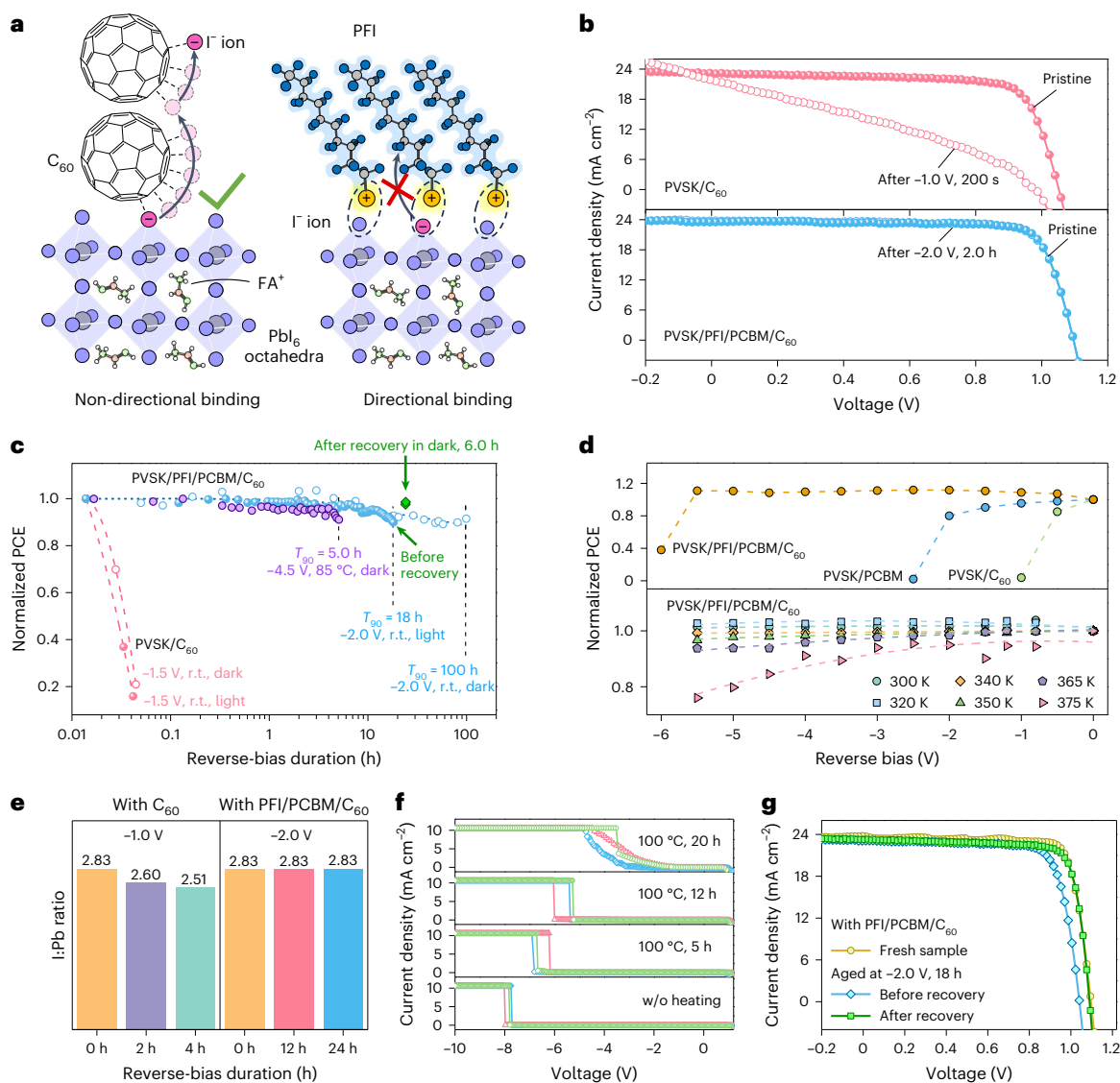


Fig. 3 | Engineered ICAPs for PSCs with improved reverse-bias stability.

a, Illustration of C_{60} with non-directional iodine affinity and PFI with directional iodine affinity (that is, a halogen bond), respectively. **b**, Comparison of current density (J)–voltage (V) curves of control PSCs (top) and PFI/PCBM/ C_{60} -based PSCs (bottom) before and after reverse-bias treatment (in situ treated under AM1.5G), respectively. **c**, Normalized PCEs of the aged PSCs (at room temperature (r.t.) or 85 °C, dark or illumination) as a function of the reverse-bias hold time. The PFI/PCBM/ C_{60} -based PSCs survived from ageing at -2.0 V under light for 18 h, and showed a self-recovery effect after a 6 h storage in the dark (green points), which survived from ageing again at -3.0 V sequentially (Supplementary Fig. 21).

d, PCE values of the aged PSCs as a function of the reverse bias applied (200 s), in which the treatments were carried out at room temperature (top) or elevated temperature (bottom) and the PCE data were extracted from J – V curves after each treatment. **e**, Changes in the iodine 3d core level peaks as a function of the reverse-bias hold time, where the XPS spectra were measured from the metal electrodes peeled off from control PSCs (bottom) or PFI/PCBM/ C_{60} -based PSCs (top), respectively. **f**, J – V curve of the PSCs with PFI/PCBM/ C_{60} interfacial layers and their perovskite films preheated at 100 °C in N_2 (before PFI coating) for different times. **g**, J – V curves of the fresh, aged (by -2.0 V, 18 h) and self-recovered (in dark, 6 h) PFI/PCBM/ C_{60} -based PSCs, respectively.

Tables 1 and 2). The device maintains 90% of its initial PCE value after an ultralong reverse-bias stress of 100 h in the dark or 18 h under sunlight at -2.0 V (Fig. 3c and Supplementary Fig. 21), with the reverse-bias stability improved by three orders of magnitude. Moreover, the PFI/PCBM/ C_{60} devices can maintain 90% of their PCE under a high bias of -4.5 V at 85 °C for 5 h (Fig. 3c,d and Supplementary Figs. 22 and 23), in which conditions the control device failed in a few seconds. Notably, this improvement cannot be achieved by replacing PFI with other interfacial layers (Supplementary Fig. 24). These are, to our knowledge, the first devices with regular thickness to survive the harsh ISOS-V-2I ageing test¹⁸.

To confirm the relationship between reverse-bias stability and the outward migration of I^-/I_x^- ions from the perovskite layer, the elements on the peeled-off silver electrodes before and after reverse-bias treatment were analysed by X-ray photoelectron spectroscopy (XPS). For control

PSCs, the peeled-off electrode showed significantly increased iodine 3d (619.11 eV and 630.52 eV) core level peaks in the first 2 h of -1.0 V treatment (Supplementary Fig. 25 and Supplementary Table 3). On the other hand, the I:Pb ratio on the exposed perovskite surface decreased from 2.83 to 2.60 (Fig. 3e, Supplementary Fig. 26 and Supplementary Table 4) due to the iodine loss. In the case of PSCs with a PFI/PCBM/ C_{60} interfacial layer, no additional iodine signal can be detected on the peeled-off electrode even after a prolonged reverse-bias treatment (-2.0 V, 24 h), and the I:Pb ratio (2.83) on the exposed perovskite surface remained unchanged.

The escape of I^-/I_x^- ions results in a gradual increase in the number of V_i defects in the perovskite layer, creating a major device failure mechanism under reverse bias. This can be demonstrated by intentionally generating V_i defects in perovskite films (before covering PFI molecules) and then carrying out the same reverse-bias stability test.

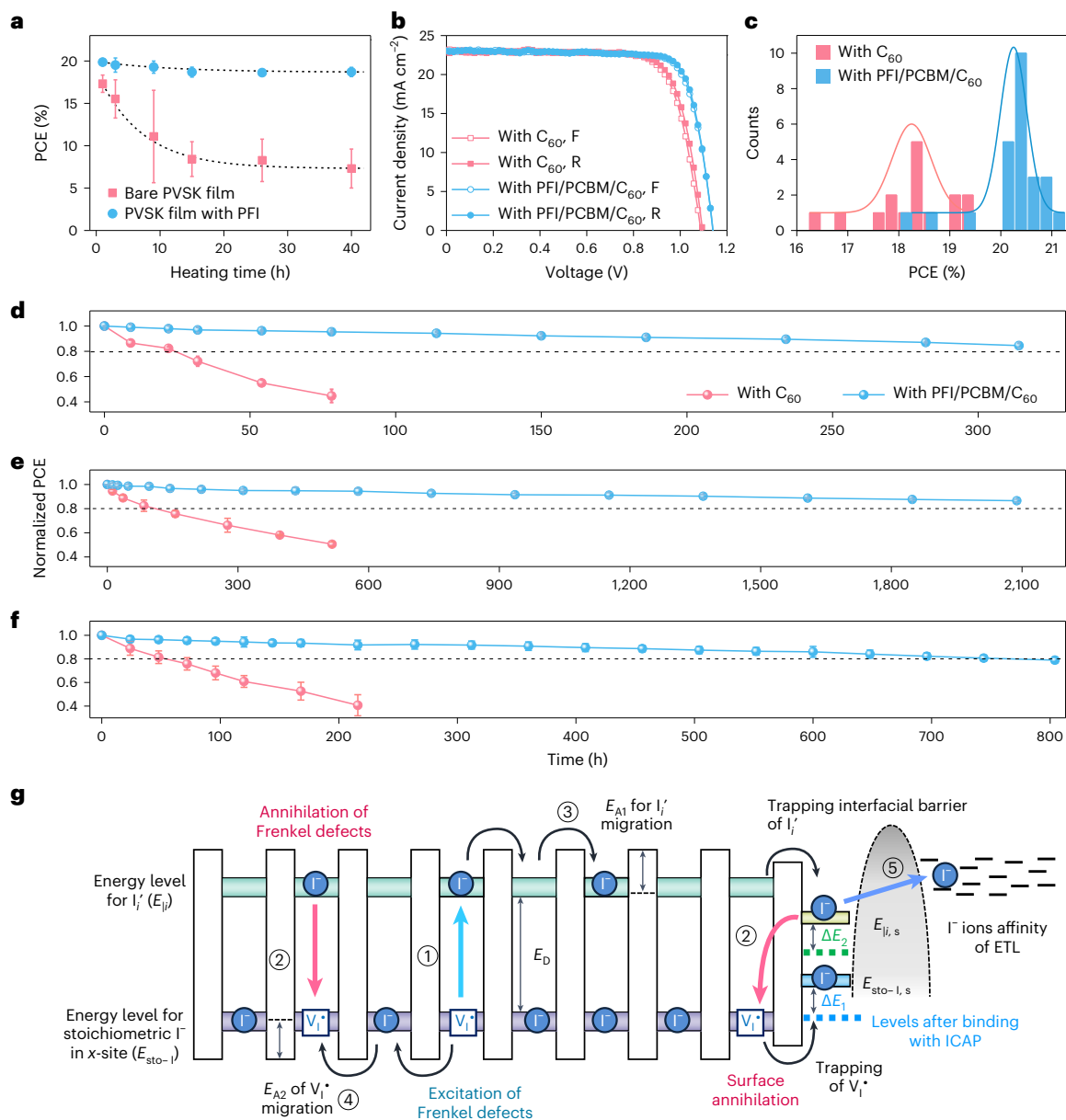


Fig. 4 | PFI/PCBM/C₆₀-based PSCs with improved efficiency and stability.

a, PCEs of PSCs fabricated with the perovskite films post-treated with light (1 sun) and thermal annealing (85 °C) for different times before the deposition of the ETL. **b**, Typical $J-V$ curves of the control and PFI/PCBM/C₆₀-based PSCs. **c**, Statistics of PCE distribution for PSCs with C₆₀ or PFI/PCBM/C₆₀ as the ETL, respectively. **d–f**, Normalized PCEs of PSCs with a C₆₀ (pink sphere) or PFI/PCBM/C₆₀ (blue sphere) interfacial layer as a function of the continuous ageing time, in which the devices were aged under ultraviolet illumination (395 nm, 60 mW cm⁻²) (**d**), 1 sun illumination (AM1.5G) at room temperature (**e**) and 1 sun illumination (LED lamp) combined with 85 °C heating (**f**). **g**, Brief illustration of the energy diagram for

the mobile defects (I_i' and V_i') in perovskites. The processes of excitation of a pair of Frenkel defects (1, with defect formation energy of E_D), bulk (left) and surface (right) annihilation between I_i' and V_i' (2), migration of I_i' (3, with activation energy of E_{A1}) and V_i' (4, with activation energy of E_{A2}), outward migration of I_i' ions from perovskites to the ETL (5) are schematically shown. Capturing the dynamically generated iodide species (I_2 and I_i') and confining iodide species inside the perovskite layer with PFI/PCBM/C₆₀ facilitates a sustainable halide circulation. Data points in **a** are presented as mean $\pm$ s.d. ($n = 10-16$). The data in **d,e** are shown as mean $\pm$ s.d. ($n = 3$). Data in **f** are presented as mean $\pm$ s.d. ($n = 20$). In cases where error bars are not visible, they are smaller than the symbol size.

Even if the escape of I_i^-/I_x^- ions can be prevented with the PFI layer, a dramatically decreased reverse breakdown voltage of PSCs from ~ -7.9 V to ~ -2.5 V can be observed once the initial concentration of V_i' defects in perovskite layer has been increased by 20 h of thermal annealing (Fig. 3f and Supplementary Note 5). Accordingly, the reverse-bias stability under sunlight at -2.0 V decreased by one order of magnitude (Supplementary Fig. 27). We speculate that the increased V_i' defects can migrate and accumulate under reverse bias, which triggers strong local band bending and charge carrier influx. For perovskite regions flooded with charge carriers, new troublesome defects would be formed or permanent destruction would occur.

The close relationship between ion migration and reverse-bias stability is also supported by studying the transient displacement current (Supplementary Note 6). In control PSCs, the displacement current caused by ion migration is significant at room temperature and can be frozen when the device is cooled to 250 K (Supplementary Figs. 28 and 29). Consistently, the control PSCs also become reverse-bias-stable at $T < 250$ K (Supplementary Fig. 30). On the contrary, no ionic displacement current can be detected in PSCs with a PFI/PCBM/C₆₀ interfacial layer up to a high reverse bias of -5.5 V (Supplementary Fig. 31).

More surprisingly, the 10% PCE loss observed in the PFI/PCBM/C₆₀-based device after 18 h of -2.0 V ageing can be self-recovered after

storage in the dark overnight (Fig. 3c,g). This self-recovery effect might be related to the gradual annihilation between the Γ^-/Γ_x^- ions accumulated at the PVSK/PFI interface and the diffusive V_i^- defects³⁶, which essentially benefits from the strong Γ^-/Γ_x^- ion confinement offered by PFI. The 1,000 times slower PCE degradation rate and the timely self-recovery effect suggest that instability related to reverse bias, or shading, is not a limiting factor for perovskite thin-film solar cells.

Solar cell performance and operational stability

The Γ^- ion confinement offered by PFI helps to prevent unintentional iodide loss during fabrication (for example, thermal annealing), which minimizes the initial V_i^- concentration in the resulting perovskite films. As a demonstration, the as-cast FACs perovskite films were intentionally annealed at 85 °C in N_2 under white light (approximating to 1.0 sun) for 40 h (Fig. 4a and Supplementary Note 5), after which a PSC was subsequently fabricated and measured. The PCE of the control sample (without PFI) dropped from 18.5% to 7.3%, whereas the PCE of the sample covered with PFI maintains 94% of its initial efficiency (19.9%). This increased heating tolerance broadens the processing windows and hence improves the fabrication yield of PSCs. Furthermore, the dipole direction of the PFI molecules on the perovskite surface is beneficial for charge collection in PSCs with an inverted structure³⁷. Due to these advantages, PFI/PCBM/ C_{60} -based PSCs generally show increased open-circuit voltage (V_{oc}) (Fig. 4b, Supplementary Fig. 32 and Supplementary Table 5) and PCE (Fig. 4c).

The operational stabilities of PSCs have been substantially improved after inserting the PFI layer. Under ultraviolet illumination (395 nm, 60 mW cm⁻²), the PFI/PCBM/ C_{60} -based device can maintain 85% of its initial PCE after 300 h of illumination, corresponding to an ultraviolet irradiation dose of 180 kWh m⁻² (Fig. 4d). The ultraviolet stability of PSCs has been improved to be 12 times of that required in the IEC61215 ultraviolet test³⁸.

When ageing near their maximum power points (MPPs) at 1 sun illumination (unencapsulated, in N_2 , AM1.5G), the PFI/PCBM/ C_{60} device maintained 90% of its initial PCE after 1,500 h, the lifetime of which has been improved by ~30 times compared with the control devices. It should be noted that PSCs are susceptible to multiple influencing factors under practical working conditions, such as interfacial degradation related to the hole transport layer. The decline in efficiency of the PFI/PCBM/ C_{60} device observed in Fig. 4e might be caused by other factors beyond the photolysis-released I_2 or the loss of Γ^- ions, which need to be further addressed by combining ICAP with other stabilizing strategies. Nevertheless, running the PSCs near the MPP condition at an elevated temperature of 85 °C can add more insights into the intrinsic stability of PFI/PCBM/ C_{60} -based devices. In contrast to the control device, which showed ~40% efficiency decline after 120 h of operation, the PFI/PCBM/ C_{60} -based device maintained 80% of its initial PCE after 765 h of operation (Fig. 4f and Supplementary Fig. 33), which is among the highest stability of inverted PSCs achieved so far under such harsh test conditions.

All these much improved stabilities described above are attributed to the PFI layer. The PFI layer dynamically scavenges the released iodine at elevated temperature while preventing iodine loss from the perovskite layer, which is a different mechanism with the steric effect on organic cations and water molecules³⁹. For a better understanding, a brief schematic energy diagram of the Γ^- ion is proposed in Fig. 4g and explained in Supplementary Note 7. The PFI has a superior role compared with C_{60} in preventing Γ^- ion loss because it eliminates the iodine transport channels near the PVSK/ETL interface, and therefore creates a closed and benign circulation of iodine-related defects in halide perovskites.

In summary, due to the long diffusion distance of I_2 in perovskites, the highly detrimental I_2 deep in the perovskite layers can be scavenged by ICAPs on the surface of perovskites. Conversely, the reverse-bias stability of inverted PSCs is very sensitive to ICAPs on the perovskite

surface. Directional halogen bonds offered by PFI molecules can lock the Γ^- ions and facilitate a sustained iodide circulation in the perovskite layer. PSCs with PFI/PCBM/ C_{60} as the ETL can survive 100 h of the ISOS-V-I ageing test (−2.0 V at room temperature) and 5 h of the ISOS-V-2I ageing test (−4.5 V at 85 °C). These devices also show comprehensively improved ultraviolet stability (12 times higher dosage than the IEC61215 requirement) and thermal-light stability (85 °C, 765 h at conditions near the MPP).

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41563-024-01876-2>.

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Methods

Materials

N,N-Dimethylformamide (DMF, 99.8%, anhydrous), dimethylsulfoxide (DMSO, 99.8%, anhydrous), chlorobenzene (CB, 99.8%, anhydrous), IPA (anhydrous, 99.8%), PbI_2 (99.999%), PbBr_2 (99.999%), CsI (99.99%), PFI (97%), TOPO (99%), PVP ($M_w = 160,000$), DIO (98%), bathocuproine (BCP, 99.99%), *N,N'*-diphenyl-3,4,9,10-perylenedicarboximide (PTCDI, 98%), 1,2,3,4,8,9,10,11,15,16,17,18,22,23,24,25-hexadecafluoro-29*H*,31*H*-phthalocyanine (F_{16}CuPc ; dye content, 80%), perylene-3,4,9,10-tetracarboxylic dianhydride (PTCDA, 97%) and 5FIB (99%) were purchased from Sigma-Aldrich. Toluene (99.5%) was purchased from Sinopharm Chemical Reagent. HTP (98%) was purchased from J&K Scientific. Fullerene (C_{60} , 99.95%) was purchased from Nano-C. Methylammonium iodide (MAI) and formamidinium iodide (FAI) were purchased from Greatcell Solar. PTAA was purchased from Xi'an Polymer Light Technology. PCBM was purchased from Luminescence Technology. All reagents and solvents were used directly if not specified.

Solution preparation

$\text{FA}_{0.90}\text{Cs}_{0.10}\text{PbI}_{2.83}\text{Br}_{0.17}$ perovskite. FAI (2.91 M), PbBr_2 (1.36 M) and PbI_2 (1.41 M) were dissolved in DMF:DMSO = 9:1 (v/v) mixed solvent, and CsI (1.50 M) was dissolved in DMSO. The 'mixed' perovskite precursor solutions of $\text{FA}_{0.90}\text{Cs}_{0.10}\text{PbI}_{2.83}\text{Br}_{0.17}$ were prepared with FAI (0.90 M), PbI_2 (0.92 M), PbBr_2 (0.085 M) and CsI (0.10 M) from the predissolved solution; this perovskite precursor solution was then aged for 6–8 h at 65 °C before use.

MAPbI₃ perovskite. For the preparation of MAPbI₃ perovskite solution, the PbI_2 solution was prepared by dissolving 1.3 M PbI_2 into 1 ml DMF solvent and stirring for 6 h before use. The MAI solution was prepared by dissolving 60 mg MAI in 1 ml IPA.

Device fabrication

The indium tin oxide glass was cleaned with detergent, deionized water, acetone, ethanol and IPA successively. PTAA was dissolved in chlorobenzene to give a 2.0 mg ml⁻¹ solution and then spin-coated on precleaned indium tin oxide substrate at 6,000 r.p.m. for 30 s and annealed at 65 °C for 10 min in N_2 .

For MAPbI₃ perovskite film, a two-step method was used. The PbI_2 solution was spin-coated on the substrate at 3,000 r.p.m. for 30 s. Next, the MAI solution was spin-coated at 3,000 r.p.m. for 30 s, and then the films were annealed at 80 °C for 1 h.

For $\text{FA}_{0.90}\text{Cs}_{0.10}\text{PbI}_{2.83}\text{Br}_{0.17}$ perovskite film, a one-step method was used. The perovskite precursor solutions of $\text{FA}_{0.90}\text{Cs}_{0.10}\text{PbI}_{2.83}\text{Br}_{0.17}$ were spin-coated at 6,000 r.p.m. for 30 s with pouring chlorobenzene as an antisolvent. Then the substrates were annealed at 100 °C for 20 min in N_2 . PCBM with a concentration of 15 mg ml⁻¹ in CB was spin-coated on perovskites at 3,000 r.p.m. for 30 s and annealed at 80 °C for 10 min.

For perovskite films coated with soft Lewis acids (for example, C_{60} , PCBM, PFI and 5FIB) or soft Lewis bases (for example, TOPO, PVP, HTP and DIO), the PFI (2.5 mg ml⁻¹ in IPA), 5FIB (8% in IPA, v/v), TOPO (5 mg ml⁻¹ in toluene), PVP (1 mg ml⁻¹ in IPA) or HTP (1% in CB, v/v) layers were fabricated by spin-coating at 4,000 r.p.m. for 30 s. C_{60} , F_{16}CuPc , PTCDI or PTCDA were evaporated in vacuum on the perovskite films at a rate of 0.1 Å s⁻¹ to give layers 10 nm thick. For the introduction of extrinsic I₂ molecules to perovskite films, predissolved iodide solution (10 mM in IPA, 80 °C for 10 h) was deposited onto the perovskite by spin-coating at 4,000 r.p.m. for 30 s, followed by a quick baking at 65 °C for 5 min.

Finally, for the BCP (7 nm thick) interfacial layer, copper or silver electrodes (80 nm thick) were thermally evaporated sequentially on the films in vacuum at a rate of 0.2 Å s⁻¹ and 2 Å s⁻¹, respectively.

Device testing

The current density (*J*)–voltage (*V*) curves of PSCs were measured in an N_2 glovebox with a Keithley 2400 at a voltage scan rate of 0.02 V s⁻¹, a delay time of 50 ms and a sweep region from −0.2 V to 1.2 V (that is, forward voltage scanning) under 100 mW cm⁻² AM1.5G illumination provided by an AAA-class solar simulator (Enli Technology). An NREL-certificated silicon reference cell (SRC-2020, Enli Technology) was used for calibration. The mismatch factor was calculated to be 0.96. The device area is defined to be 0.10 cm² by metal masks. The external quantum efficiency (EQE) was characterized by a QE-R solar cell quantum efficiency measurement system (Enli Technology), and the light source was a 75 W xenon lamp. The monochromatic light intensity for the EQE was calibrated with a NIST-certified silicon photodiode from 300 nm to 1,100 nm. The EQE spectrum was integrated over AM1.5G photon flux to obtain the photocurrent density. In the probe station chamber, under a vacuum of 10⁻³ Pa, the ion activation energy of a lateral solar cell was measured. A Keithley 4200 semiconductor analyser was used for voltage–current characterization. The displacement current of PSCs under reverse bias was measured with a Keithley 4200 semiconductor analyser.

Device operational and electrical bias stability tests

Long-term operational stability measurements of unencapsulated perovskite devices were operated under a solar simulator (Newport 94011A) with a light intensity of 100 mW cm⁻² AM1.5G in N_2 . The temperature of the devices under illumination was measured to be 40–45 °C. All devices were loaded with a resistor so that they worked at conditions near the MPP during the test. The light and thermal stability of unencapsulated PSCs were measured under a 1 sun condition (white LED lamp; the light intensity was adjusted to maintain fresh PSCs with a_{JSC} of 23 mA cm⁻²) combined with 85 °C heating in N_2 . The ultraviolet light ageing test of the perovskite devices was operated under a lamp with 395 nm and 60 mW cm⁻² in N_2 . The reverse-bias stability test for the PSCs was performed in the N_2 glovebox, where a specific reverse bias (for example, −0.5 V to −5.5 V) was provided by the Keithley 2400, and the *J*–*V* curve of the device was measured immediately after a given bias was applied to the device for a certain time.

Film characterization

The chemical state and surface composition of the samples were analysed by XPS using an ESCAKAB 250Xi system equipped with a monochromatic SPECS XR-MF X-ray source (Al K α , $h\nu = 1,486.7$ eV; base pressure, 2.0×10^{-10} mbar). Raman scattering measurements on perovskite films were taken with an inVia Qontor system (Renishaw) using 532 nm excitations with a spot size <10 μm and an 1,800 lines mm⁻¹ grating. SEM images of the perovskite crystals were obtained by using a scanning electron microscope (TESCAN MIRA3 LMU) equipped with an electron beam accelerated at 10–20 kV and a vacuum transfer accessory at KW-ST lab (www.kewei-scitech.com). The absorption spectra were obtained by using an ultraviolet–visible spectrometer (Thermo Evolution 201) in the spectral range of 300–1,100 nm.

NMR spectra

The ¹⁹F NMR measurements were performed by a Bruker AVANCE-500 instrument.

Ion migration studies

The activation energy (E_a) for ion migration was measured at different temperatures with a Keithley 4200 source meter using a lateral device with the structure gold/perovskite/gold on a glass substrate. The electric field of the lateral device was 0.2 V μm^{-1} . The device was put in a Lakeshore Probe Station to obtain the desired temperature.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding authors on reasonable request. Source data are provided with this paper.

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Author contributions

Y. Yuan conducted the project, conceived the ideas, designed the experiments and analysed the data. J. Huang supervised the project and analysed the data. J. He supervised the Raman spectral and absorption spectral studies. X.R. fabricated the perovskite films, and

studied the iodide/polyiodide capture effect of Lewis acids. J.W. and X.R. fabricated the solar cells, and obtained the $J-V$ curves, the EQE and the NMR spectra. X.R. and J.W. measured the activation energies of ion migration, the reverse-bias stability and the working stability of solar cells. X.R., Y.L. and Y.W. carried out the ultraviolet-visible absorption spectrum measurements, X.R. and B.Y. carried out the SEM measurements, J.W., H.X. and Y.G. carried out the XPS test and analysis. X.R. and H.H. carried out the Raman spectral tests. Y. Yan carried out the density functional theory calculations. Y. Yuan, J. Huang and J. He wrote the paper; all authors revised the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41563-024-01876-2>.

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

Manuscript device testing, page 21

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

Manuscript device testing, page 21

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☒ Yes
☐ No

Manuscript Fig. 4b

Voltage scan conditions

☒ Yes
☐ No

Manuscript device testing, page 21

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

Manuscript device testing, page 21

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes
☒ No

Provide a description of the protocol.

This study focuses on device stability. Without any bias or light preconditioning.

Stability of the J-V characteristic

☒ Yes
☐ No

Manuscript Fig. 4d-4g

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☒ Yes
☐ No

We observed hysteretic behavior for the aged devices, hence we test the J-V curve with both forward and backward direction.

Explain why this information is not reported/not relevant.

Related experimental data

☒ Yes
☐ No

Manuscript Fig. 4b

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

Supplementary Fig. 32

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator

☒ Yes
☐ No

Manuscript Fig. 4b and Supplementary Fig. 32d

Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	Provide a description of the measurement conditions. N/A
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Manuscript device testing, page 21 Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Manuscript device testing, page 21 Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	Manuscript device testing, page 21 Explain why this information is not reported/not relevant.
6. Mask/aperture		
Size of the mask/aperture used during testing	☐ Yes ☒ No	Report the size of the mask/aperture. This study focuses on device stability. No mask was used in this study.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	Report the difference in the short-circuit current density values measured with the mask and aperture area. No mask was used in this study.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☐ Yes ☒ No	Identify the independent certification laboratory. N/A. This study focuses on device stability.
A copy of any certificate(s)	☐ Yes ☒ No	Certificate copies should be provided in the Supplementary information. Please state the supplementary item number. N/A. This study focuses on device stability.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	Manuscript Fig. 4, Supplementary Fig. 32, Supplementary Fig. 33 Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	Manuscript Fig. 4, Supplementary Fig. 32, Supplementary Fig. 33 Explain why this information is not reported/not relevant.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Manuscript device testing, page 21-22, Manuscript Fig. 4, Supplementary Fig. 21, Fig. 23, Fig. 24, Fig. 33 Explain why this information is not reported/not relevant.